

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	SWTID-2026-3830
Project Name	EduGenie: Google Gemini Powered learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation used in all files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static, models and utilities.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names such as user_data, input_text used.
4	Function / Method Names	Functions are descriptively named	Yes	Verb-based names such as load_data(), predict_result().
5	Code Comments	Inline and block comments explain logic	Partial	Key functions commented; a few sections need more comments.
6	Modular Design	Code is split into reusable functions/modules	Yes	Logic divided into separate functions and modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused imports and duplicate code removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	try-except used for input and file handling.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas, NumPy)	Training and prediction pipeline	High
2	Prediction Function	Python (Scikit-learn)	Web app and testing scripts	High
3	Input Validation Helper	Python	All form/input handling	Medium